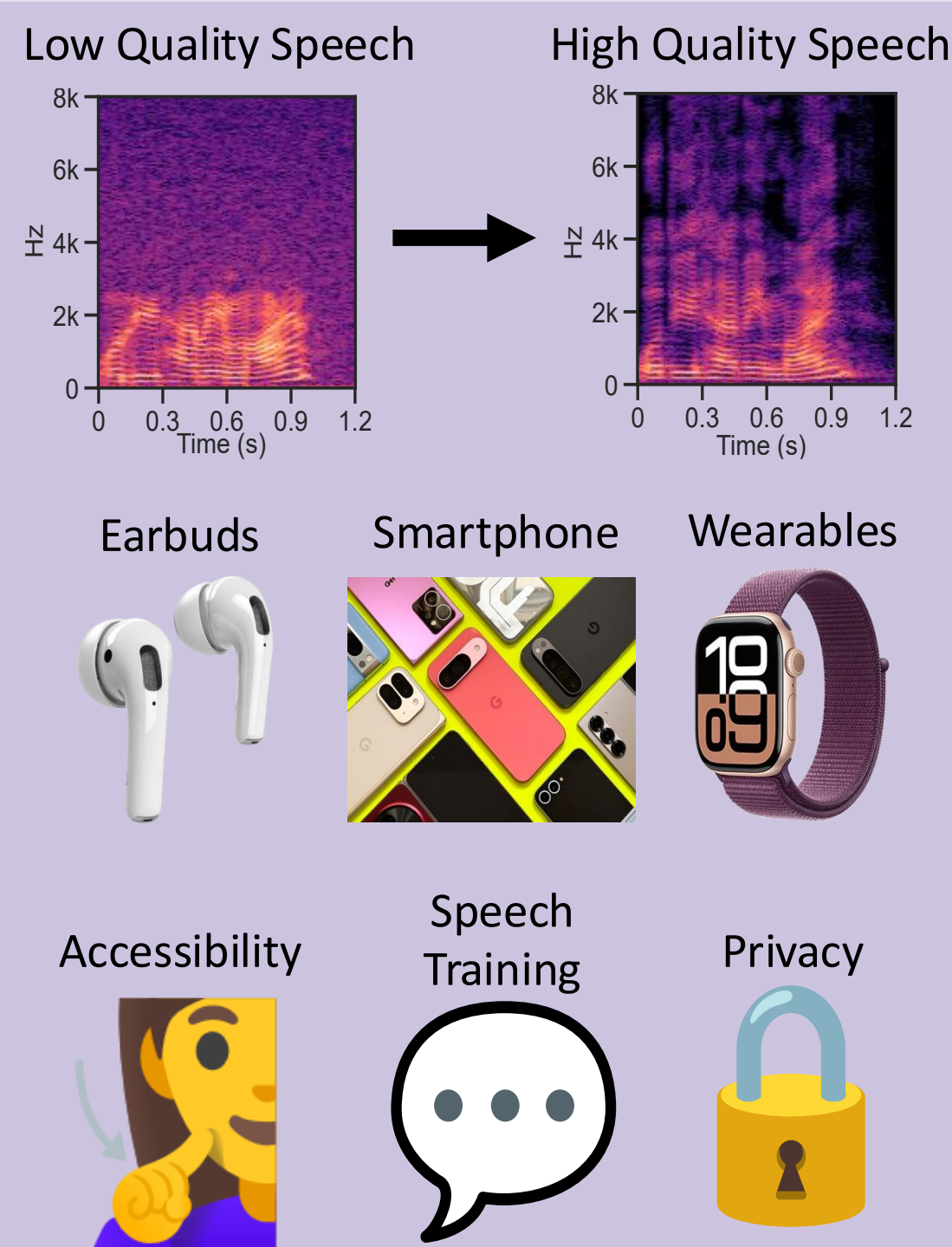


Motivation

• **Motivation:** Speech enhancement and super resolution improves voice quality, clarity, intelligibility, and reduces noise.

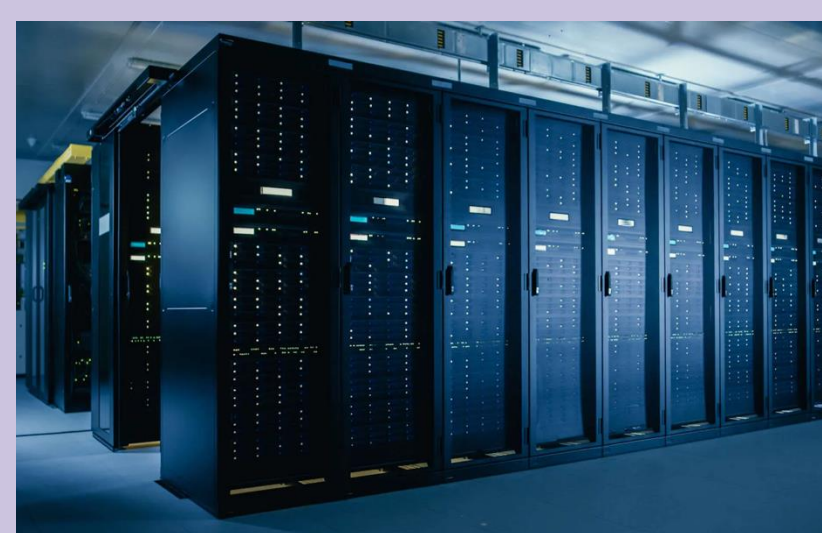
• **Vision:** Enable real-time and high-fidelity super resolution for mobile and resource-constrained platforms for audio and vibration sensing

• **Broader Impacts:** high quality voice on-the-go, accessibility, on-device privacy, speech (impediment) training

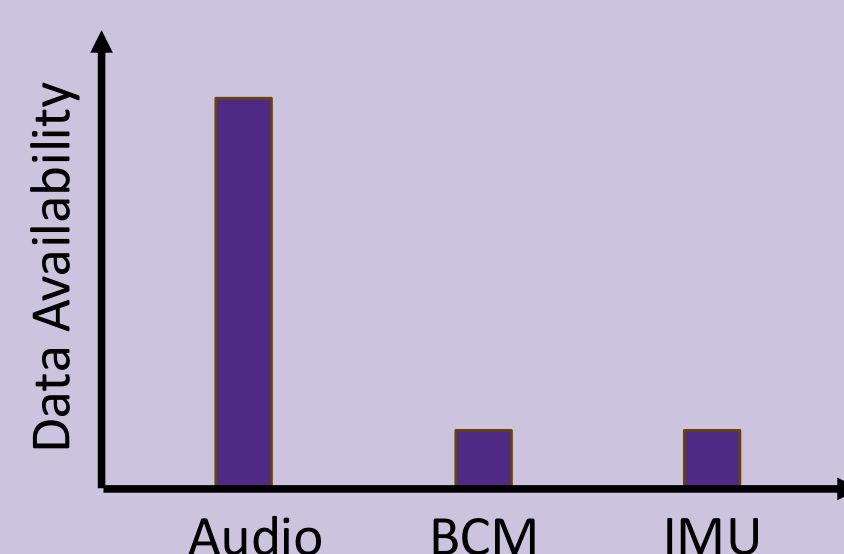


Limitations with State-of-Art Methods

Heavy Processing Required

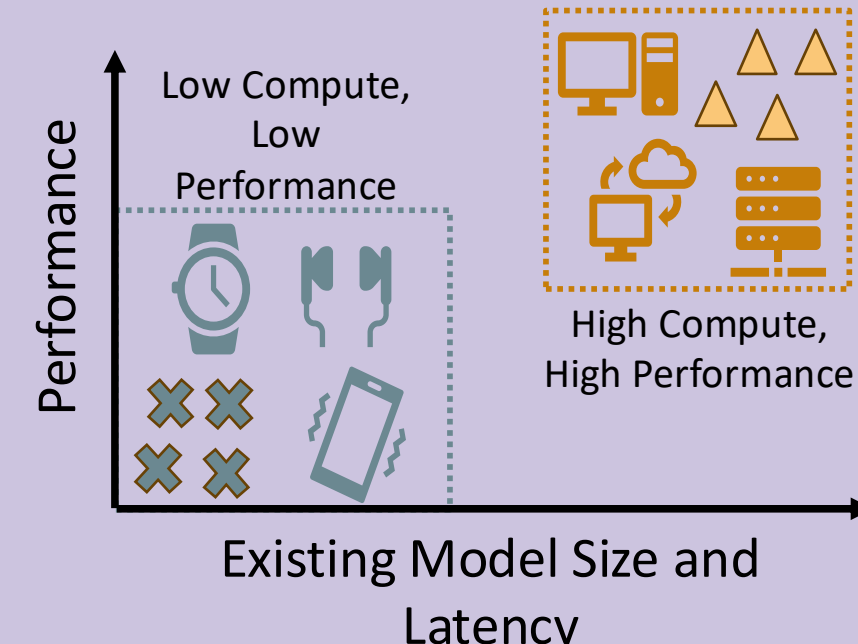


Data Scarcity

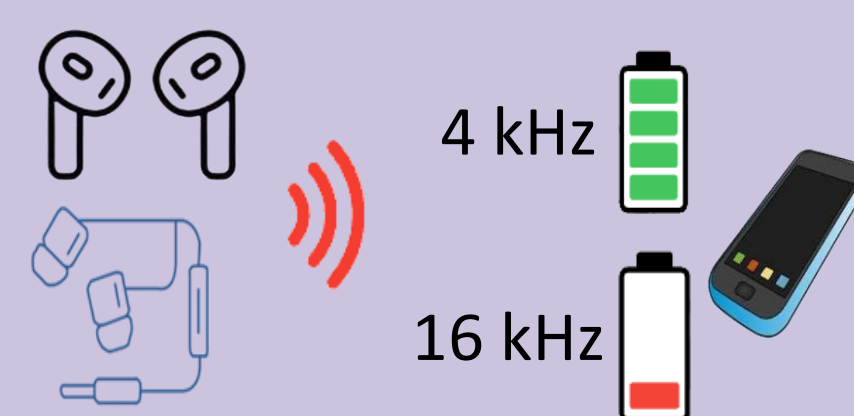


Performance Gap

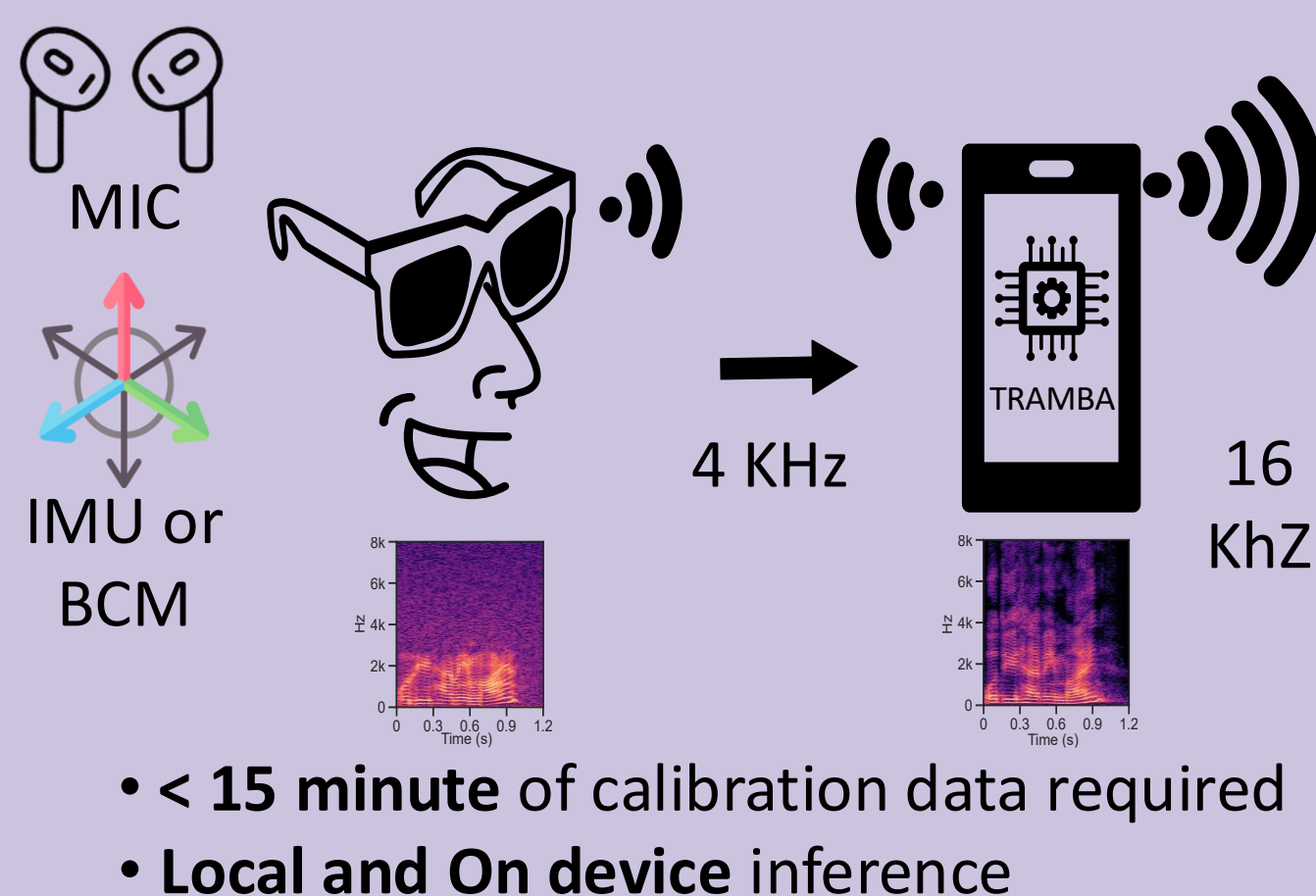
Goal: Low Compute, High Performance



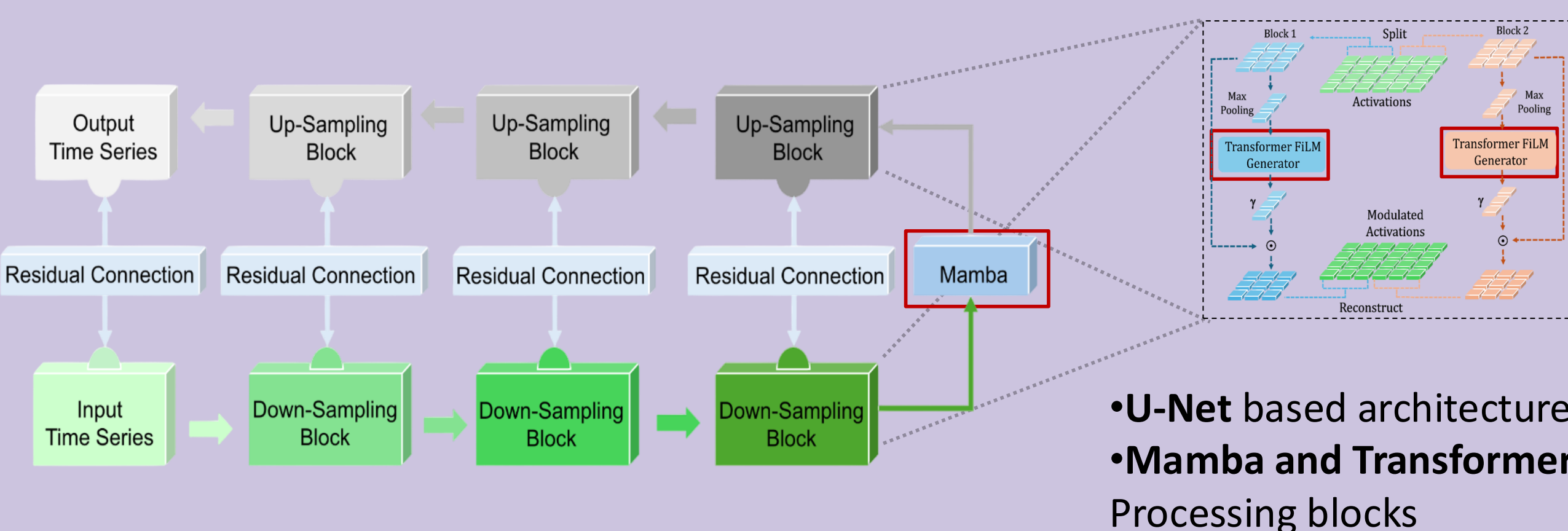
System Optimizations



TRAMBA Mobile Processing Pipeline

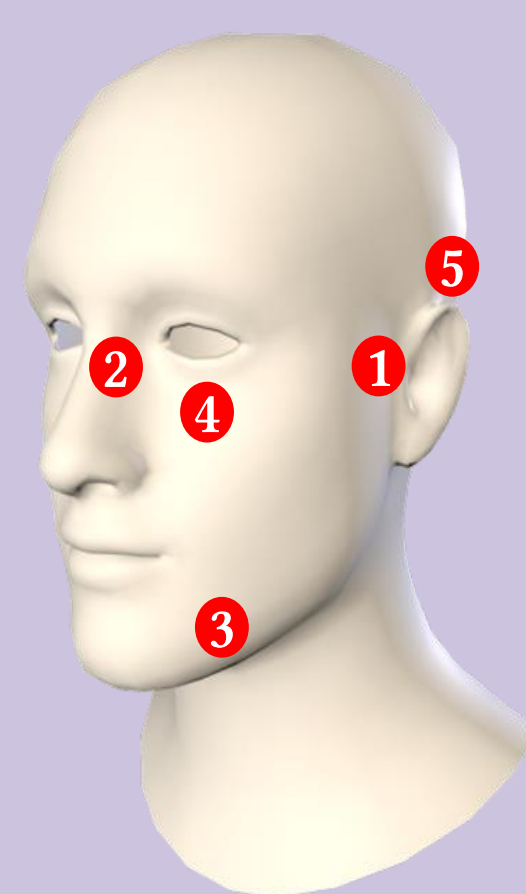
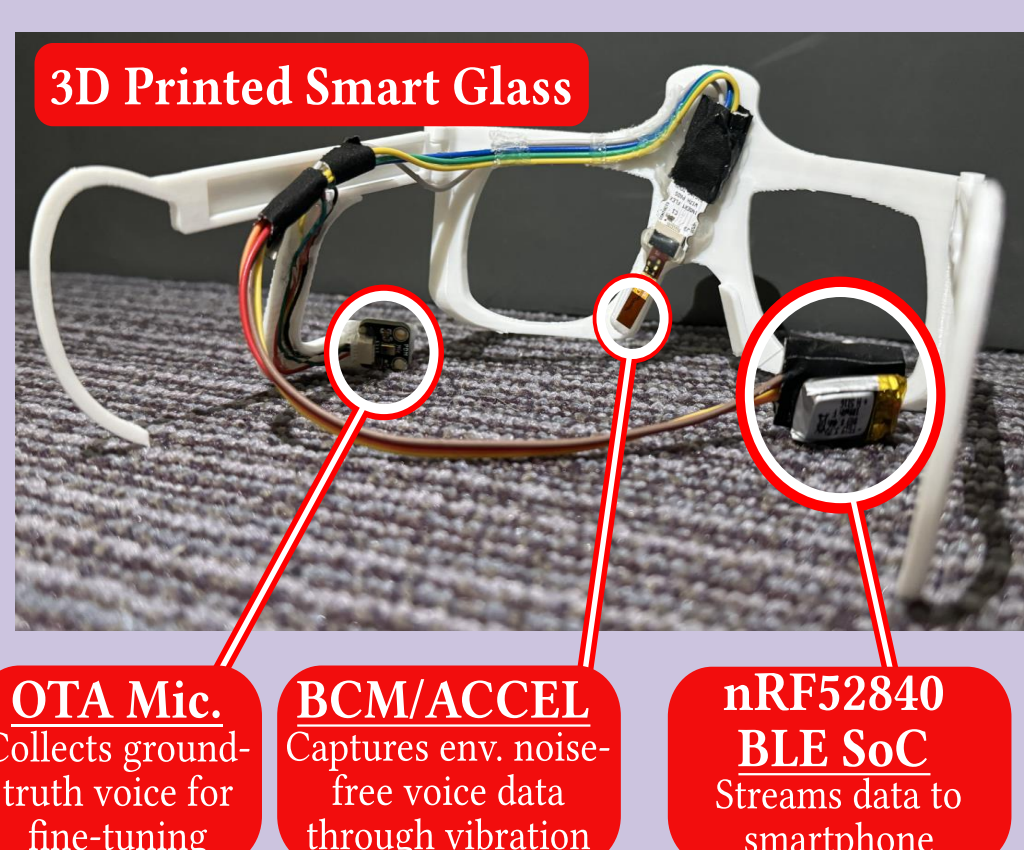


TRAMBA Super Resolution Architecture

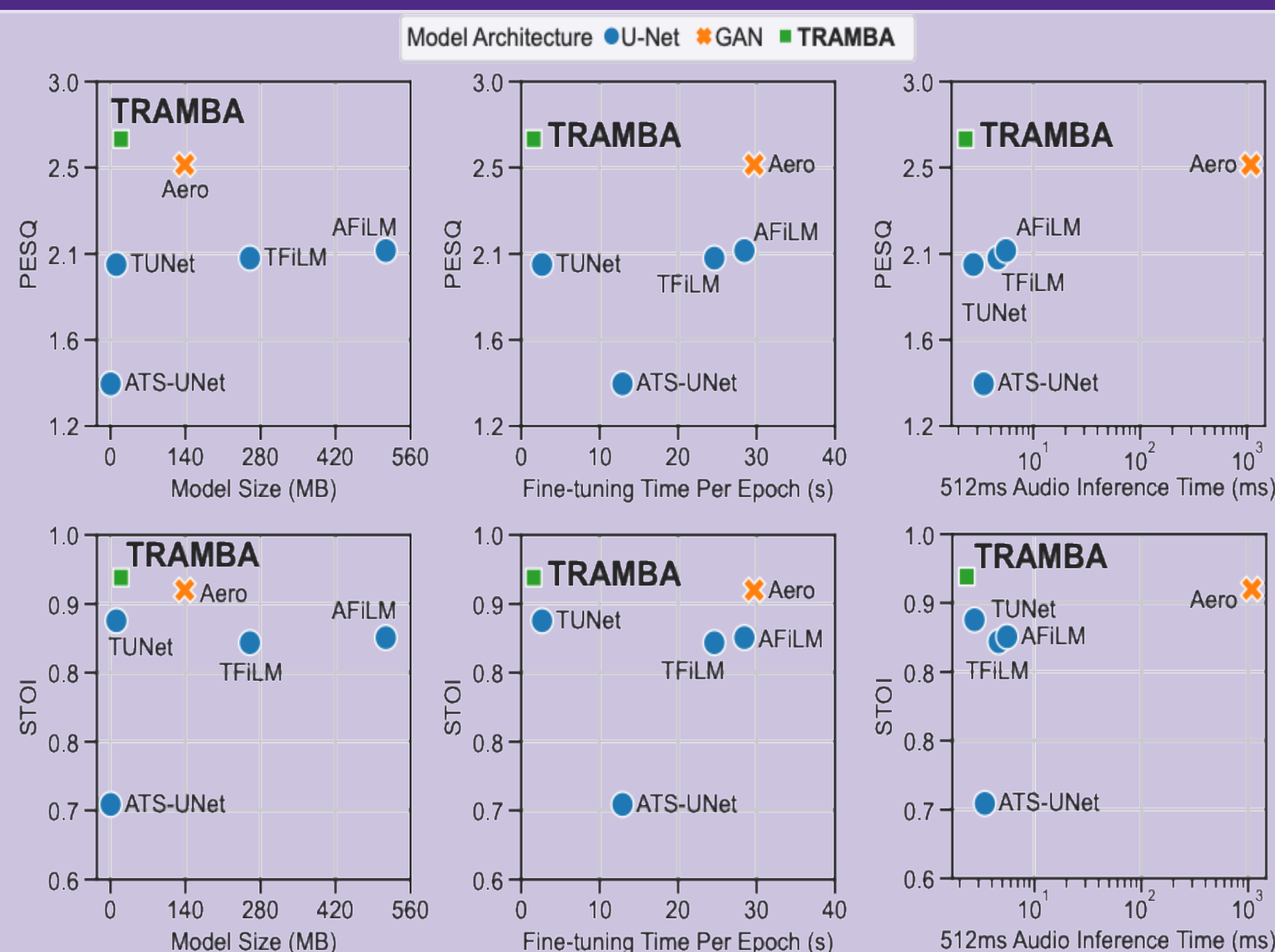


Evaluation Platform, Settings, and Results

- Glasses wearable platform for prototyping and evaluation
- Five locations on face
- Four environments
- Four user states:
 - Standing still
 - Walking
 - Jogging
 - Running



- 1 Temporal Bone (Bone-conduction Headsets)
- 2 Nasal Bone (Smart Glasses)
- 3 Mandible (Face Masks)
- 4 Zygomatic Bone (VR Headsets)
- 5 Parietal Bone (Hats and Bone-conduction Headsets)



- Experiments on L40 GPUs
- Comparisons against State-of-art models including GANs
- Achieves the **highest PESQ** (perceptual quality) and **STOI** (speech intelligibility) scores, and comparable **model size**, **fine-tuning time** and **latency**

Power and Latency

	Data Rate	Power Consumption
TRAMBA- 500 Hz	8 kbps	2.49 mW
TRAMBA- 1 kHz	16 kbps	2.58 mW
TRAMBA- 2 kHz	32 kbps	2.75 mW
TRAMBA- 4 kHz	64 kbps	3.21 mW
TRAMBA- 8 kHz	128 kbps	4.09 mW
No processing (16 kHz)	256 kbps	6.48 mW

- 20 ms processing time for 512 ms of audio on common phones
- Power reduction by ~50%

References

[1] Birnbaum, et al. *Advances in Neural Information Processing Systems* 32 (2019) [2] Rakotonirina, et al. *2021 IEEE 31st International Workshop on Machine Learning for Signal Processing (MLSP)*. [3] Nguyen, et al. "ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)". [4] Li, Yuang, et al. *Sensors* 23.1 (2022): 35. [5] Mandel, et al. *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023.

Conclusion

- **Outperforms state-of-art GAN** approaches by up to 7.3% in PESQ and 1.8% in STOI while requiring up to **400x less latency** and **an order of magnitude less memory**
- **Even greater performance boosts over U-Net** architectures by up to 40% in PESQ and 10% STOI with **comparable compute**
- **Real-time processing on resource-constrained platforms** with **~50% power reduction** compared to conventional sampling
- TRAMBA is a critical step for high fidelity speech enhancement on any platform